

Please note:

- The duration of the exam is three hours. Question paper has 2 pages and 9 questions.
- No source material is to be used.
- Go through every question carefully before answering. Please write down all the steps in your solutions.

Maximum Marks: 70

1. Let C be an (n, k) binary linear code. Prove the following:

(a) If $\mathbf{x} \notin C^\perp$, then $\sum_{\mathbf{y} \in C} (-1)^{x_1 y_1 + x_2 y_2 + \dots + x_n y_n} = 0$

(b) Prove that $\sum_{\mathbf{x} \in C} w_H(\mathbf{x}) \leq n2^{k-1}$.

$$\begin{aligned} x &= (x_1, x_2, \dots, x_n) \\ y &= (y_1, y_2, \dots, y_n) \end{aligned}$$

(4 marks)

(3 marks)

2. Given a (n_1, k_1, d_1) code C_1 and a (n_2, k_2, d_2) code C_2 , the direct product of C_1 and C_2 denoted by $C_1 \otimes C_2$ is an $(n_1 n_2, k_1 k_2, d)$ code constructed as follows:

Arrange a message of length $k_2 k_1$ in a matrix of size $k_2 \times k_1$ by filling the matrix entries one row after another. Let this matrix be denoted by M . Encode each row of M by the code to obtain an intermediate matrix of size $k_2 \times n_1$. Encode each column of this intermediary matrix with code C_2 to obtain a matrix of size $n_2 \times n_1$. Rearrange this matrix into a vector, by reading the entries row after row. Now, answer the following questions:

(a) Prove that the distance of the code is $d_1 d_2$.

(5 marks)

(b) Show that the code matrix can be obtained from the message matrix M using the following matrix map: $C = G_2 M G_1^T$, where M is of size $k_2 \times k_1$ with entries from $\mathbb{F}_q$.

(4 marks)

3. Construct a standard array for the ternary (field is $\mathbb{F}_3$) linear code with the generator matrix

$$C = G_2^T H G_1$$

$$G = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Use this array to decode $[1, 2, 1, 1]$.

(5 marks)

(b) Construct the syndrome table for the above code.

(4 marks)

4. State and prove the Plotkin bound for codes. Compute an upper bound on the number of codewords in a double-error-correcting binary linear code of length 8 based on this bound.

(5 + 3 marks)

5. A burst of length ℓ is the event of having errors in a codeword such that locations i and j of the first (leftmost) and last (rightmost) errors, respectively, satisfy $j - i = \ell - 1$. Let C be a linear $[n, k > 0]$ over $\mathbb{F}_q$ and suppose that there exists a decoder for C that corrects every burst of length ℓ or less. Then, answer the following:

(a) Show that in every nonzero codeword $\mathbf{c} \in C$ the locations i and j of the first and last nonzero entries in C must satisfy $j - i \geq 2t$. (2 marks)

(b) Show that $n - k \geq 2t$. (3 marks)

6. (a) Let $\alpha_1, \alpha_2, \dots, \alpha_q$ be elements of $\mathbb{F}_q$ and let r be an integer in the range $1 \leq r < q - 1$. Show that

$$\sum_{j=1}^q \alpha_j^r = 0.$$

(b) Find the inverse of $x^2 + 1$ in $\mathbb{F}_3[x]/(x^3 + 2x + 1)$. Please find it systematically (and not via guess work). (4 marks)

7. Consider $\mathbb{F}_2$ and suppose the finite field with 16 elements Φ is represented as $\mathbb{F}_2[x]/(x^4 + x^3 + x^2 + x + 1)$. Let β be a root of $x^4 + x^3 + x^2 + x + 1$ in Φ . Express all the elements of Φ in terms of β . Is β primitive in Φ ? Show that $\beta + 1$ is a primitive element of Φ . (5 marks)

8. Consider the $n \times n$ Vandermonde matrix

$$X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{n-1} & \alpha_2^{n-1} & \dots & \alpha_n^{n-1} \end{bmatrix}.$$

$$(XB)_{i,j} = \sum_{\ell=1}^n$$

For $\ell = 1, 2, \dots, n$, let $b_\ell(x)$ be the polynomial over $\mathbb{F}$ defined as

$$b_\ell(x) = b_{\ell,0} + b_{\ell,1}x + \dots + b_{\ell,n-1}x^{n-1} = \prod_{j \neq \ell} \frac{x - \alpha_j}{\alpha_\ell - \alpha_j},$$

$$b_1(x) = \prod_{j \neq 1} \frac{x - \alpha_j}{\alpha_1 - \alpha_j}$$

i.e., $b_\ell(x)$ is the unique $(n-1)$ -degree polynomial such that

$$b_\ell(\alpha_j) = \begin{cases} 1, & j = \ell, \\ 0, & \text{otherwise.} \end{cases}$$

$$b_1(\alpha_1) = \frac{\alpha_1 - \alpha_1}{\alpha_1 - \alpha_1}$$

Let B be the $n \times n$ matrix given by

$$B = \begin{bmatrix} b_{1,0} & b_{1,1} & \dots & b_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n,0} & b_{n,1} & \dots & b_{n,n-1} \end{bmatrix}.$$

(a) Show that $B = X^{-1}$.

(b) Let C be an $[n, k]$ Reed-Solomon code over $\mathbb{F}$ with generator matrix G whose $(i, j)^{\text{th}}$ entry is α_j^{i-1} , for $1 \leq i \leq k$, $1 \leq j \leq n$, and $\alpha_1, \alpha_2, \dots, \alpha_n$ being distinct elements in $\mathbb{F}$. (2 marks)

Let $\bar{G} = [I|A]$ be a systematic generator matrix of C . Show that the $(i, j)^{\text{th}}$ entry in A is given by

$$\prod_{s \in [k]: s \neq i} \frac{\alpha_j + \alpha_s - \alpha_s}{\alpha_i - \alpha_s},$$

for $1 \leq i \leq k$, $1 \leq j \leq n - k$.

9. (a) Prove that if the generator polynomial of a binary cyclic code satisfies $g(1) = 0$, then all codewords of C have even weight. (4 marks)

(b) Let C be a cyclic code of length n over $\mathbb{F}_q$ with generator polynomial $g(x)$, where $g(x) | (x^n - 1)$. Let $h(x)$ be such that $g(x)h(x) = x^n - 1$. Then, prove that $C^\perp$ is a cyclic code with generator polynomial $x^{2h}(x^{-1})$. (3 marks)

(6 marks)